Word2vec

Serry Sibaee
Research Engineer (ANLP) - Riotu Labs

Semantic space concept

Semantic (meaning) converted to a vector (embedding)

$$\mathcal{D} = \{d_1, d_2, \ldots, d_n\}$$

$$f : \mathcal{D} \to \mathbb{R}^d$$

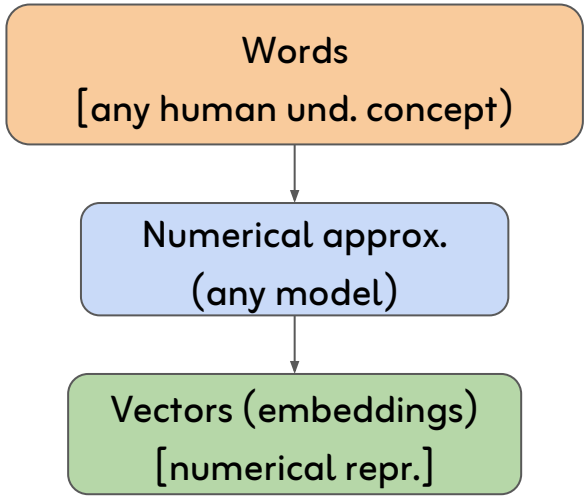
$$\mathbf{v}_i = f(d_i)$$

$$\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$$

$$s : \mathbb{R}^d \times \mathbb{R}^d \to [0, 1]$$

$$s(\mathbf{v}_i, \mathbf{v}_j)$$

$$\|f(d_i) - T(d_i)\| \leq \epsilon, \quad \forall d_i \in \mathcal{D}$$



Vectorizing concepts

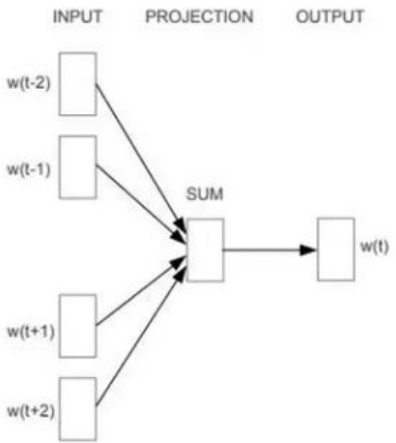
1. Counting methods: **Bag of Words** [Parse]
2. Statistical:
    a. Simple: TF-IDF ... etc [Parse]
    b. Complex (depends on neurals usually) [dense]
        i. Word2vec and its derivatives (fast-text, GloVe ... etc) without context awareness
        ii. BERT and transformers based (with context awareness)

Note: the difference between parse and dense is that: **parse is understandable (countaing) embedding and very huge R^n** while **dense is compact inhuman conceptualize embedding. R^d**

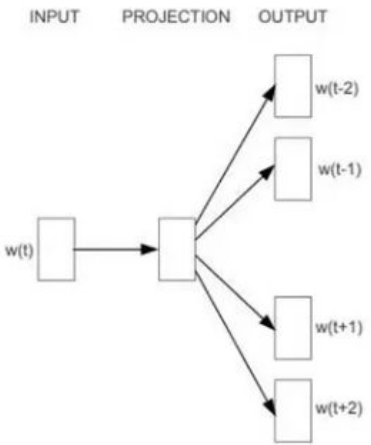
n >> d

Word2vec method

A neural network method that trains a model to predict the hidden word/s from the context. (more details link or [the book]) with two main methods: CBOW (counting bag of words) or Skip gram.



CBOW



Skip-gram

link

Math foundation ⟶ w2v [link, link2]

3.1 Skip-Gram Model

Given a corpus with vocabulary V, the Skip-Gram model aims to maximize the probability of context words w_O given a target word w_I:

$$J = \frac{1}{T}\sum_{t=1}^{T}\sum_{-c\leq j\leq c, j\neq 0} \log P(w_{t+j}|w_t) \tag{1}$$

where c is the context window size. The conditional probability is defined using softmax:

$$P(w_O|w_I) = \frac{\exp(\mathbf{v}_{w_O}^\top \mathbf{u}_{w_I})}{\sum_{w\in V}\exp(\mathbf{v}_w^\top \mathbf{u}_{w_I})} \tag{2}$$

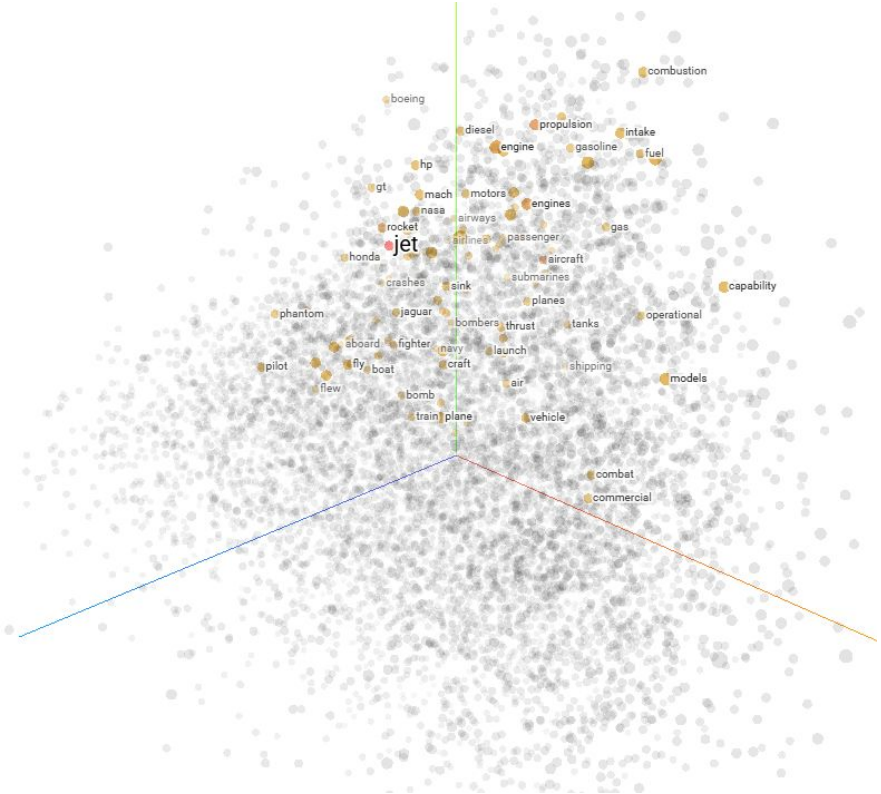
where $\mathbf{u}_{w_I}$ and $\mathbf{v}_{w_O}$ are the input and output word embeddings, respectively.

3.2 CBOW Model

CBOW predicts the target word given the sum of its surrounding context word embeddings:

$$P(w_t|w_{t-c},...,w_{t+c}) = \frac{\exp(\mathbf{v}_{w_t}^\top \sum_{j=-c}^{c}\mathbf{u}_{w_{t+j}})}{\sum_{w\in V}\exp(\mathbf{v}_w^\top \sum_{j=-c}^{c}\mathbf{u}_{w_{t+j}})} \tag{3}$$

Space of meanings (semantics)



link

Look at this book

Sentence Embeddings

We have for each word one embedding but for a full sentence?

how to use word2vec with sentences

Summation

$$\mathbf{v}_S = \sum_{i=1}^{n} \mathbf{v}(w_i)$$

Averaging

$$\mathbf{v}_S = \frac{1}{n} \sum_{i=1}^{n} \mathbf{v}(w_i)$$

Weighted Sum

$$\mathbf{v}_S = \sum_{i=1}^{n} \alpha_i \mathbf{v}(w_i), \quad \sum \alpha_i = 1$$

Train a classification model

Training Flow Titles:

1. Load Pre Trained Word2Vec Model [use AraVec or others]
2. Convert Text to Word Embeddings
3. Prepare the Dataset [from hugging face]
   or other ideas for classification (like language, toxicity, ... etc)
4. Split Data into Training and Testing Sets
5. Train the Logistic Regression Model
6. Evaluate Model Performance
7. Prediction Function
8. Test a Sample Prediction

